

Upper Respiratory Culture, Routine

Test Details

METHODOLOGY

Culture

RESULT TURNAROUND TIME

2 - 4 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Culture; isolation of potential aerobic pathogens, identification (additional charges/CPT code[s] may apply), and susceptibility testing if culture results warrant (additional charges/CPT code[s] may apply). CPT coding for microbiology and virology procedures often cannot be determined before the culture is performed. Requests with only a written order and no test number indicated will be processed according to **Default Testing for Routine Microbiology** (/node/258).

Use

Isolate and identify potentially pathogenic organisms from throat, sinus etc; evaluate pharyngitis; evaluate nares for staph

Special Instructions

Sterile specimens (eg, sinus aspirates or tympanocentesis fluid), collected by invasive procedures, should be submitted as sterile body fluid cultures. If the amount of material aspirated is small, it may be advisable to inject it into an anaerobic transport or absorb it onto the swab of the bacterial swab collection kit and use the bacterial transport. For throat specimens submitted for isolation of *Neisseria gonorrhoeae*, use GC (Neisseria gonorrhoeae) Culture Only [008128] (/tests/008128/gc-i-neisseria-gonorrhoeae-i-culture-only) and include inoculated Jembec® transport. Specimens from other sources, such as genital, stool, urine, upper and lower respiratory specimens, cannot be cultured under the aerobic bacterial culture test number. If specimens are incorrectly submitted with an order for aerobic bacterial culture, the laboratory will process the specimen for the test based on the source listed on the test request form. The client will not be telephoned to approve this change, but the change will be indicated on the report. Check expiration date of transport; do **not** use expired device.

Limitations

Interpretation requires a significant level of experience and technical proficiency in order to avoid false-positives and false-negatives.¹ Many other etiologic agents can be responsible for pharyngitis.² **Note:** This procedure does not include screening for *Neisseria gonorrhoeae* or *Corynebacterium diphtheriae*. Anaerobic organisms that are frequently implicated in chronic infection of the tonsils and adenoids are not recovered by aerobic culture methods.

Custom Additional Information

Thrush, oral candidiasis, and *Candida* esophagitis frequently complicate antineoplastic therapy, hyperalimentation, transplantation immunosuppression, pregnancy, and the acquired immunodeficiency syndrome (AIDS). In addition to a fungal culture, a saline wet preparation, Gram stain, or KOH preparation demonstrating yeast cells or pseudohyphae may also be useful in rapidly establishing the diagnosis of oral or mucocutaneous candidiasis.

***Streptococcus pyogenes*:** (group A β -hemolytic strep) and other β -hemolytic streptococci in groups B, C, D, and G are generally susceptible to penicillin and its derivatives, therefore, susceptibility need not be routinely determined. The principal reason for considering an alternative drug for individual patients is allergy to penicillin. Erythromycin, a cephalosporin, or clindamycin might be substituted in these cases. Patients allergic to penicillins may also be allergic to cephalosporins.

In the late 1980s a resurgence of serious *Streptococcus pyogenes* infection was observed. Complications including rheumatic fever, sepsis, severe soft tissue invasion, and toxic shock-like syndrome (TSLs) are reported to be most common with the M1 serotype and a unique invasive clone has become the predominant cause of severe streptococcal infections.³

Ear: Normal flora of the skin of the healthy ear includes *Staphylococcus epidermidis*, *Corynebacterium* sp, and *Staphylococcus aureus*. Correlation of nasopharyngeal cultures with results of tympanocentesis culture is poor and lacks predictive value in identification of the causative agent of otitis media. In decreasing order of frequency, the following organisms have been recovered from tympanocentesis: *S pneumoniae* (50% to 75%), *H influenzae* (10% to 30%), *Moraxella (Branhamella) catarrhalis* (5% to 10%), *Streptococcus pyogenes* (5% to 10%), *Staphylococcus aureus* (1% to 5%), *Pseudomonas aeruginosa* (0.1% to 1%). *E coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* may be isolated from neonates. In therapeutic failures, *S aureus*, and *P aeruginosa* are most frequently recovered. Tympanocentesis is not usually performed in primary infections. It is to be considered in treatment failures and neonates. *Candida* superinfection may complicate therapy for recurring ear infections and may be a cause of persistent otorrhea. Otitis externa is frequently caused by *P aeruginosa* and less frequently by *Candida* sp, *Proteus* sp, *S aureus*, and *Trichophyton* sp.

Specimen Requirements

Specimen

Throat swab, nasopharynx swab, nares swab, ear swab

Volume

One swab or aspirated material

Container

Bacterial culture transport swab

Collection Instructions

Throat: Depress tongue and rub swab vigorously over each tonsillar area and posterior pharynx. Any exudate should be touched, and care should be taken to avoid the tongue and uvula. Place swab in transport.

Nasopharynx: With patient's head immobilized, insert flexible wire swab into nostril until it reaches posterior nares. Leave swab in place for 15 to 30 seconds. Rotate and remove. Place swab in transport.

Reference Range

Routine respiratory flora present or no growth

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Unlabeled specimen or name discrepancy between specimen and request label; inappropriate specimen transport device; leaking specimen; specimen received after prolonged delay (usually more than 48 hours); expired transport

References

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Footnotes

1. Bibler MR, Rouan GW. Cryptogenic group A streptococcal bacteremia: Experience at an urban general hospital and review of the literature. *Rev Infect Dis*. 1986 Nov-Dec; 8(6):941-951. PubMed 3541128 (<http://www.ncbi.nlm.nih.gov/pubmed/3541128>)
2. Lang SD, Singh K. The sore throat. When to investigate and when to prescribe. *Drugs*. 1990 Dec; 40(6):854-862. PubMed 2079000 (<http://www.ncbi.nlm.nih.gov/pubmed/2079000>)
3. Cleary PP, Kaplan EL, Handley JP, et al. Clonal basis for resurgence of serious *Streptococcus pyogenes* disease in the 1980s. *Lancet*. 1992 Feb 29; 339(8792):518-521. PubMed 1346879 (<http://www.ncbi.nlm.nih.gov/pubmed/1346879>)